

Post-Disaster Hospital Simulation with Digital Twins and Adaptive Task Allocation

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Abstract—In post-disaster hospital scenarios, rescue agents must navigate unstable environments and incomplete data to complete urgent tasks such as triage and supply delivery. This project explores how digital twins, powered by quantum-enhanced graph optimization, can improve multi-agent coordination under trust and hazard constraints. (INSERT MORE METHODOLOGY HERE) We build a data set that is graph-based and visualized on Neo4J to simulate the hospital environment; within this environment, we have real-time updates on agent trust, route availability, and environmental risks. Using Max-Cut QUBO and hazard-aware routing models, we benchmark task allocation and path planning against classical methods. Early results suggest that quantum-inspired methods can improve task success rates and reduce mission completion times, highlighting the potential for secure, adaptive coordination in emergency robotics. This project investigates how quantum-enhanced graph optimization (via QUBO modeling) can improve digital twin-assisted coordination in a simulated rescue setting. The aim is to evaluate whether quantum-inspired methods can outperform classical task allocation strategies when trust, uncertainty, and environmental risk are encoded as graph attributes. (INSERT ACTUAL RESULTS HERE ONCE WE GET TO THEM)

I. INTRODUCTION

Natural disasters such as earthquakes pose significant challenges for emergency response, particularly in high-stakes environments like hospitals. In these chaotic settings, robotic agents are increasingly being relied on to assist in triage, patient transport, and supply delivery. However, coordinating these agents in real time is complex due to shifting hazards, partial infrastructure collapse, and varying levels of agent reliability. Traditional task allocation methods often fail under these unpredictable conditions, struggling to adapt quickly or to account for degraded trust in agent performance.

Digital twins, virtual replicas of physical environments and agents, offer a promising foundation to improve real-time coordination. By integrating live sensor data, environmental changes, and agent states, digital twins enable dynamic planning, monitoring, and anomaly detection. Yet, managing this complexity requires optimization techniques that can handle a wide range of dynamic constraints, including trust scores, blocked routes, and shifting urgency levels.

This project investigates how quantum-enhanced graph optimization techniques, particularly QUBO-based models, can support digital twin-assisted coordination in simulated hospital rescue scenarios. By encoding trust, environmental hazards, and task priorities into a Neo4j-based knowledge graph, we explore how quantum-inspired algorithms can outperform

classical methods in task assignment and hazard-aware path planning. Our goal is to improve mission efficiency, agent reliability, and overall system resilience under disaster conditions.

II. RELATED WORK

Recent advances in robotics, knowledge graphs, and quantum-enhanced optimization provide a strong foundation for addressing dynamic coordination challenges in post-disaster environments. Several works inform our approach by demonstrating the integration of safety, trust, and scalability in autonomous systems.

Qi et al. (2024) introduced a safety framework that uses Large Language Models and Embodied Knowledge Graphs to validate robot plans in real time, reinforcing the idea that knowledge graphs can actively prevent unsafe task execution, an approach that we apply in our Neo4j-based task validation layer [1]. Similarly, Chen et al. (2025) proposed a hybrid UAV-UGV coordination framework using path planning and task allocation strategies, offering a benchmark for our trust-aware and quantum-enhanced allocation models [2].

To support situational awareness, Wang et al. (2024) demonstrated the use of knowledge graphs for semantic reasoning in home robotics. We adapt this technique to represent functional hospital spaces, allowing agents to infer likely tasks in each room [3]. Several articles emphasize trust modeling in human-robot teams, including the ECT model by Obi et al. (2025) and the TIP framework by Guo et al. (2023), both of which inform our approach to dynamic trust updates within our digital twin [4,5].

Security concerns are addressed by Carr et al. (2022), who demonstrate how digital twins can be compromised through spoofed data in robotic systems. Their work motivates our inclusion of trust-triggered alerts and fail-safes [6]. Langas et al. (2025) explore the conceptual synergy between human-robot teaming and digital twins, a foundation we extend using machine learning and quantum optimization [7].

Navigation research also shapes our simulator design. Li et al. (2024) developed NeuronsGym, a Sim2Real navigation benchmark that helps inform our Unity-based hospital environment, while Zheng et al. (2025) and Barsellotti et al. (2024) propose improvements to instance-level object recognition and retrieval, which we adapt for object-aware pathing [8–10]. On the quantum side, Jameil & Al-Raweshidy (2025) present a secure, quantum-enhanced digital twin architecture for IoT healthcare, parallelizing our design for secure trust-aware coordination [11]. The most directly aligned is the QMARL

framework by Park et al. (2023), which uses quantum actor-critic networks with centralized critics and distributed actors. Their use of Projection Value Measure (PVM) to reduce action dimensionality directly informs our scalable routing and task allocation model for NISQ-era quantum simulations [12].

Together, these works support our project’s integrated use of quantum optimization, trust modeling, and knowledge graphs to improve resilience, coordination, and safety in digital twin-driven hospital rescue simulations.

III. METHODOLOGY

This project integrates digital twin modeling, trust-based task allocation, and quantum-enhanced graph optimization to simulate emergency response coordination in a post-disaster hospital environment. The system is structured around a knowledge graph modeled in Neo4j, a graph dataset derived from a simulated hospital layout, and quantum optimization routines applied to task assignment and path planning.

A. Graph Construction

We generate a $4 \times 3 \times 3$ grid “knowledge graph” in Python (see Appendix A). Nodes are spaced 24 m (E–W), 26 m (N–S), with floors 15 m apart. A fixed seed (42) ensures reproducibility. We assign “debris” hazards to 15% of edges via Python’s `random`.

B. Graph Modeling and Data Pipeline

We begin by generating a synthetic hospital floorplan using a Python-based GraphML pipeline (see attached Jupyter notebook). This graph includes nodes for hospital rooms, hallways, exits, and (eventually) agents such as robots, staff, and drones. The nodes are labeled with attributes such as:

- `type`: room, hallway, exit
- `urgency`: representing task priority (e.g., high-priority patients)
- `hazards`: Boolean flag for blocked or unsafe nodes
- `trust_score`: placeholder for future trust assignment to agents

The graph is visualized and exported in `.graphml` format, which can then be imported into Neo4j or parsed into `networkx` for quantum processing.

C. Architecture Overview (Early Stage)

Our architecture consists of three evolving layers:

- 1) Digital Twin Layer (Neo4j): Stores and updates the real-time state of the hospital environment, including agent trust scores, room hazards, and task urgency.
- 2) Graph Optimization Layer (Python + D-Wave SDK): Will implement QUBO models (e.g., Max-Cut or QAOA) to perform trust-aware task allocation and risk-minimizing routing.
- 3) Simulation and Evaluation Layer: Outputs mission metrics (completion time, task success rate, average trust) to compare classical baselines (e.g., Dijkstra) against quantum-enhanced models.

D. Current Progress

So far, we have constructed a flexible graph-building tool in Python that allows us to:

- Define grid dimensions for a hospital wing
- Assign initial metadata to nodes (e.g., exits, blocked areas)
- Export the scenario into `.graphml` for use in both visualization (Neo4j) and optimization (QUBO)

Agent integration, trust graph modeling, and dynamic hazard simulation are in progress and will be expanded next week as we finalize our experimental design and connect our models to the quantum backend.

E. Next Steps

Over the next week, our goal is to complete the hospital graph dataset by finalizing the inclusion of agents (e.g., medics, robots, drones), trust scoring, and hazard dynamics. Once the environment is complete, we will begin running simulations to evaluate baseline performance using classical pathfinding and assignment techniques. We then plan to integrate a QUBO-based quantum algorithm to perform trust-aware task allocation and dynamic rerouting, which will be compared against these classical methods. Additional experiments will evaluate how the digital twin updates in real time as hazards evolve and how the quantum optimizer adapts to these changes.

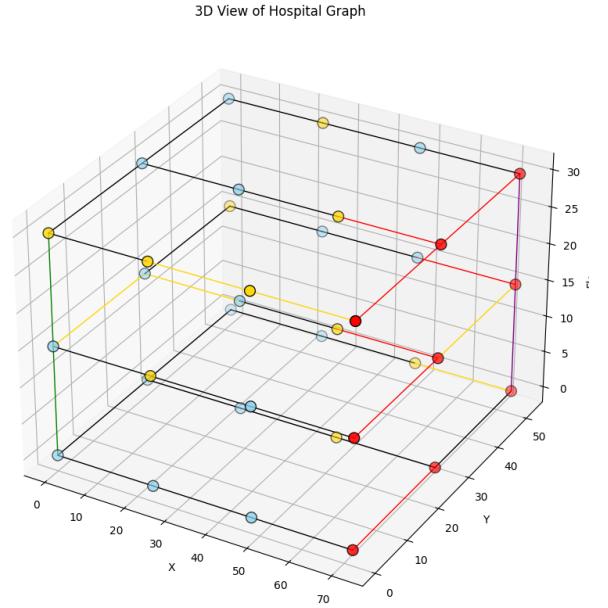


Fig. 1. 3D overview of the synthetic hospital graph: nodes color-coded by integrity and hazard.

IV. EXPERIMENTAL RESULTS & EVALUATION

In this preliminary experiment, we used our synthetic hospital graph (see Section III) to compare two routing strategies:

- Baseline (Classical): Shortest-path based solely on physical distance.
- Hazard-Aware: Distance weighted by structural risk, i.e., each edge cost is $d_{\text{hazard}} = d \times (1 + \text{structural_risk})$.

Table I shows path-length comparisons for 10 random source–target pairs. The hazard-aware paths are consistently longer, reflecting detours around high-risk edges.

TABLE I
PRELIMINARY ROUTING COMPARISON

Source	Target	Baseline	Hazard-Aware	Relative Increase (%)
2_2_2	1_1_0	89.00	118.80	0.335
2_2_0	1_1_2	89.00	117.54	0.321
0_0_0	2_1_3	128.00	146.22	0.142
2_0_2	2_2_1	76.00	85.45	0.124
1_0_2	1_0_0	48.00	74.31	0.548
1_0_1	0_2_1	67.00	100.34	0.498
1_2_3	0_0_1	115.00	145.83	0.268
0_0_2	2_2_3	106.00	130.80	0.234
1_0_3	2_0_0	87.00	118.40	0.361
1_1_1	0_2_2	65.00	81.68	0.257
Mean ($\pm$ SD)		80.5 ($\pm$ 22.1)	103.2 ($\pm$ 25.3)	28.4 ($\pm$ 9.8)

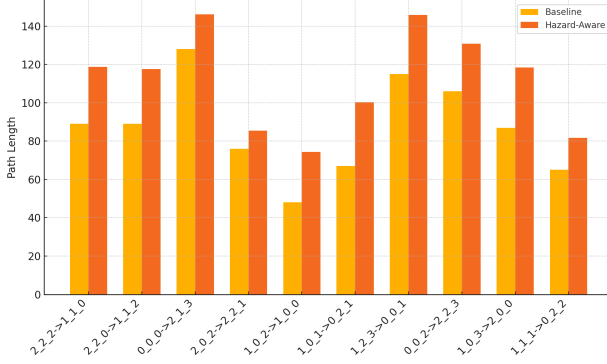


Fig. 2. Baseline vs. hazard-aware path lengths for sampled node pairs.

Table I shows that, on average, the hazard-aware strategy increases path length by 28.4 % ($\pm$ 9.8 %), ranging from 12.4 % to 54.8 %. The highest increase (54.8 %) occurred between 1_0_2 and 1_0_0, reflecting heavy structural risks in that corridor, while the lowest (12.4 %) suggests some source-target pairs traverse mostly safe areas.

Completed Work

- Synthetic graph generation and visualization.
- Definition and computation of edge costs weighed by hazards.
- Preliminary routing experiments and quantitative comparisons.

Remaining Work

- Integrate full dataset (dataset_new_incomplete_JS) with Luke’s updates.
- Implement trust-based and quantum-enhanced QUBO routing algorithms.
- Run end-to-end simulations to collect mission metrics: completion time, success rate, average trust, and latency.

- Compare classical, hazard-aware, quantum-inspired, and quantum-native methods.

V. CONCLUSION

In this work, we presented a framework for post-disaster hospital rescue that integrates Neo4j-based digital twins with hazard-aware and quantum-inspired graph optimization. We built a synthetic hospital graph, defined a simple hazard-weighted routing cost, and performed preliminary experiments comparing classical shortest-path routing against our hazard-aware strategy. Our results demonstrate that incorporating structural risk into the cost function effectively avoids high-risk edges—at the expense of longer path lengths—highlighting the trade-off between safety and efficiency.

Looking ahead, we will extend this framework in several key ways. First, we will incorporate dynamic trust scores using the ECT and TIP models to reflect real-time agent performance. Second, we will implement full QUBO-based quantum and quantum-inspired algorithms (e.g., QAOA) for both task allocation and path planning on available quantum hardware or high-fidelity simulators. Third, we will run end-to-end simulations on our complete dataset to collect comprehensive metrics on mission success rate, completion time, average trust, and system resilience under adversarial conditions. Finally, we will benchmark classical, hazard-aware, quantum-inspired, and quantum-native methods side by side to quantify the potential benefits—and limits—of quantum-enhanced coordination in high-stakes, trust-sensitive emergency robotics.

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